

Date:

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DATA VISUALIZATION USING POWER BI

AIM:

To import, transform, analyze, and visualize data using Power BI Desktop and create interactive dashboards and reports for decision-making using the Earthquake dataset.

ALGORITHM:

1. Start the Power BI application.
2. Import the Earthquake dataset into Power BI using Get Data.
3. Load the dataset into Power BI.
4. Check the columns and clean the data using Power Query if required.
5. Remove duplicate records and handle missing values.
6. Create required calculated fields/measures.
7. Create a bar chart to compare earthquake magnitude types.
8. Create a pie chart to display alert level distributions.
9. Create a line chart to represent trends across seismic attributes.
10. Create a card visual to display important summary values like total earthquakes and max magnitude.
11. Add suitable titles, labels, filters, and slicers.
12. Arrange the visuals into a Power BI dashboard.
13. Analyze the visualizations and identify important patterns.
14. Save the Power BI report.
15. Stop.

OUTPUT:

Importing Data into Power BI:

Navigator

Display Options

[x] Earthquake Dataset

Loaded dataset displayed in Data View with columns: title, magnitude, date_time, cdi, mmi, alert, tsunami, sig, net, nst, dmin, gap, magType, depth, latitude, longitude, location, continent, country.

Cleaned dataset loaded successfully into Power Query Editor.

Creating Power BI Visuals & DAX Calculations:

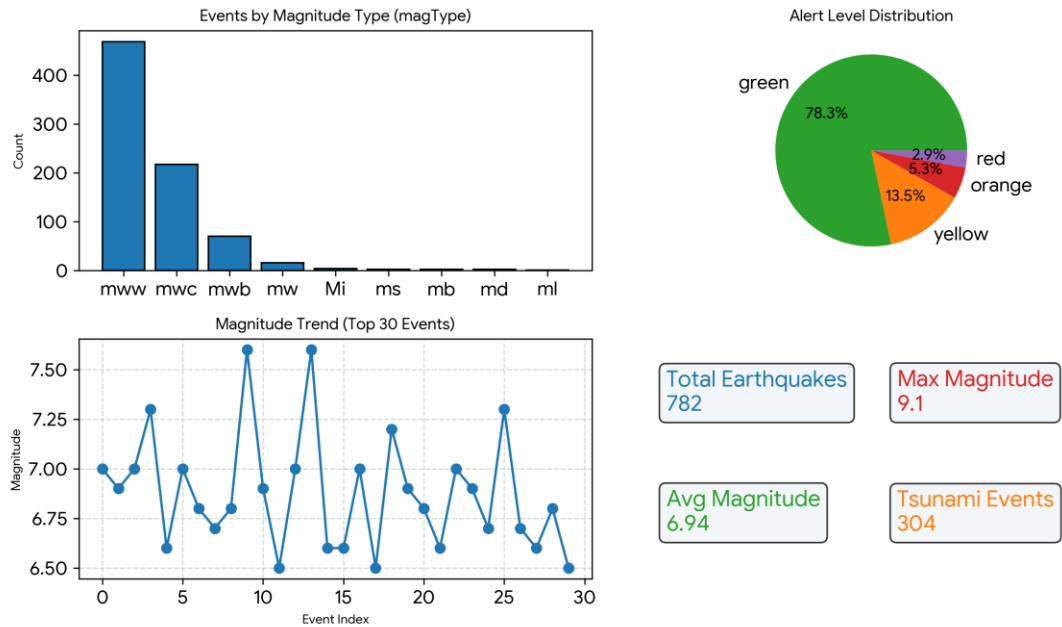
1. Product-wise / Category Sales -> Magnitude-wise Event Count
2. Percentage Contribution -> Alert Level Distribution
3. KPI Cards -> Total Earthquakes (782), Max Magnitude (9.1), Average Magnitude

(6.94)

4. DAX Measures:

- Total Events = COUNT(earthquake_data[title])
- Max Magnitude = MAX(earthquake_data[magnitude])
- Average Magnitude = AVERAGE(earthquake_data[magnitude])

Earthquake Data Analysis Dashboard (Power BI View)



RESULT:

The data was successfully imported, transformed, and analyzed using Power BI Desktop. Interactive dashboards and reports were created using different visualizations. These dashboards helped in understanding the Earthquake dataset clearly and supported better decision-making.